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Explicit Dynamics Report – Deep Drawing

(Nonlinear Finite Element Analysis - SBANGP6320EN)

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1. Introduction: Deep Drawing

1.1. Principle

Figure (1.1.1) shows the main parts of the deep drawing process, which is die, blank (part), blank holder, and punch. Also showing the reaction forces affecting on the blank. Figure (1.1.2) shows the process of deep drawing as the blank is turning into cylindrical cup.

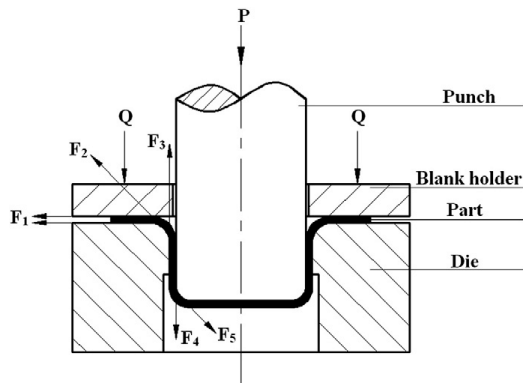


Fig. (1.1.1)

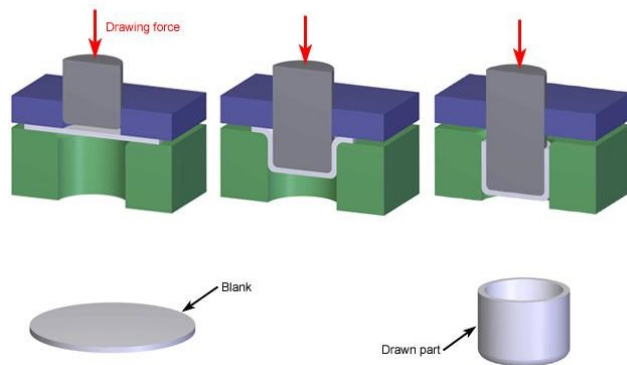


Fig. (1.1.2)

1.2. Defects

Figure (1.2.1) shows possible defects for a deep drawing process while manufacturing a cylindrical cup. There are many parameters which might affect the result of the deep drawing such as the punching force, fixturing of the blank holder, problem of the die design, and so on.

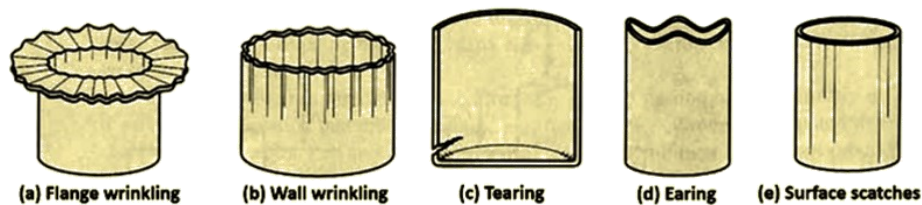


Fig. (1.2.1)

2. Model

The model could be done in 2D axis symmetric or as a 3D geometry. The 3D Model was chosen as it allows for defective simulation results not only the proper process, which can be considered more realistic. Figure (2.1) shows the Geometrical model which is used in the simulation.

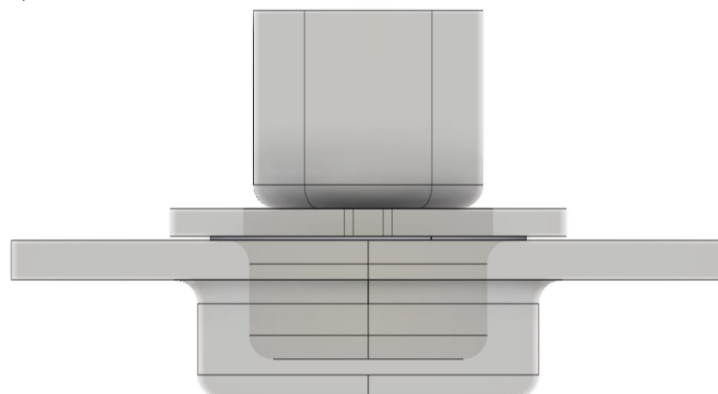


Fig. (2.1)

3. Materials

3.1. Material Model

Defining material properties can significantly change the results. The properties selection should be based on the goal and the type of simulation. In this case from the standard material libraries in Ansys: “General Non-linear Materials”, aluminum and structural steel was selected. And from “Explicit Materials”, aluminum was selected. Tables (01) and (02) show the main properties of the selected materials.

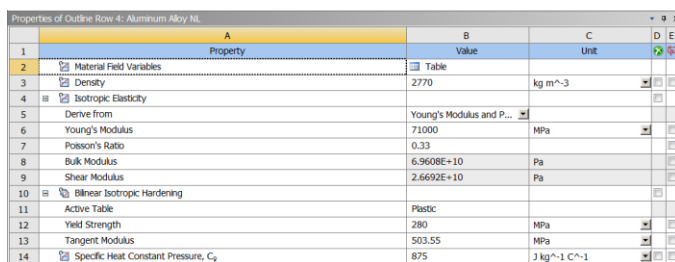
General Non-Linear Materials	Isotropic Elasticity		Bilinear Isotropic Hardening		Density [kg/m ³]
	Young's Modulus [MPa]	Poisson's ratio [1]	Yield Strength [MPa]	Tangent Modulus [MPa]	
Aluminum	71e3	0.33	280	503.55	2770
Structural Steel	200e3	0.3	250	1450	7850

Table (01)

General Explicit Materials	Bilinear Isotropic Hardening		Shock EOS Linear			Density [kg/m ³]
	Yield Strength [MPa]	Shear Modulus [MPa]	Gruneisen Coefficient	C1 [m/s]	S1 [1]	
Aluminum	280	503.55	2.1	5380	1.337	2770

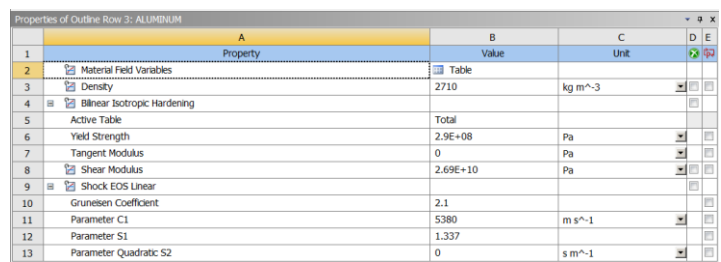
Table (02)

To understand the main difference between nonlinear and explicit aluminum, it is clearly shown in the defined properties as the nonlinear aluminum has “Isotropic Elasticity”, and “Bilinear Isotropic Hardening” with an active table which can be defined as “plastic” as shown in fig (3.1.1) and this can be extended by adding a failure method for the explicit simulation like “plastic strain failure”. While the Explicit Aluminum contains bilinear isotropic hardening also, it does not contain the tangent modulus value as shown in fig (3.1.2) and its active table can’t be modified to plastic without adding “Isotropic Elasticity”, but the main point here is that if the active table was changed to plastic the “Shock EOS Linear” will be deactivated and the material will lose its failure method.



Property	Value	Unit
Density	2770	kg m ⁻³
Young's Modulus	71000	MPa
Poisson's Ratio	0.33	
Bulk Modulus	6.9608E+10	Pa
Shear Modulus	2.6692E+10	Pa
Yield Strength	280	MPa
Tangent Modulus	503.55	MPa
Specific Heat Constant Pressure, C _p	875	J kg ⁻¹ C ⁻¹

Fig. (3.1.1)



Property	Value	Unit
Density	2710	kg m ⁻³
Yield Strength	2.9E+08	Pa
Tangent Modulus	0	Pa
Shear Modulus	2.69E+10	Pa
Gruneisen Coefficient	2.1	
Parameter C1	5380	m s ⁻¹
Parameter S1	1.337	
Parameter Quadratic S2	0	s m ⁻¹

Fig. (3.1.2)

4. Contacts

4.1. Body Interaction

Body interaction shown in fig (4.1.1) is an automatic contact tool in Ansys Explicit Dynamics scoped to all bodies. The contacts will be identified automatically between the bodies by the solver on initialization.

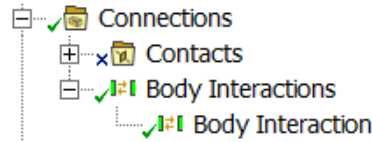


Fig. (4.1.1)

4.2. Contact Settings

The contact settings inside the body interaction were set to frictional instead of bonded with friction coefficient of 0.19 [1] which is an average value for aluminum and steel contact. The dynamic friction coefficient was set to 0.1 [1].

5. Initial Conditions

5.1. Velocity

Initial velocity of 200 m/s is applied to the punch in the negative direction of y axis and the end time of the simulation is calculated using the formula (02) by providing the traveled distance d [mm] and the velocity v [mm/s]. The velocity in the software is not a real physical velocity but it represents the velocity of spreading information in the mesh. Therefore, it can significantly control the simulation time “processing time” as it defines the end time of the simulation.

$$v = 200 \left[\frac{m}{s} \right] = 200e3 \left[\frac{mm}{s} \right] \quad (01)$$

$$t = \frac{d \left[\frac{mm}{s} \right]}{v \left[\frac{mm}{s} \right]} = \frac{38}{200000} = 0.00019 \text{ [s]} \quad (02)$$

5.2. Analysis Settings

The analysis settings contain the main controls of the simulation, for example sub-stepping options and number of calculated points during the simulation, etc. Table (03) shows the entered values. The end time together with the velocity can represent the end point of the simulation, for example in this case the maximum position of the punch.

Number Of Steps	1
Load Step Type	Explicit Time Integration
End Time	0.00019
Output Controls → Save Results On	Equally spaced Points
Output Controls → Result Number of Points	100

Table (03)

6. Boundary Conditions

6.1. Punch:

A remote displacement was set to the die to restrict its rotation and translation except for y axis. The used values are shown in table (04) and the applied remote displacement is shown in fig (6.1.1).

Type	Value	Type	Value
X component	0	Rotation X	0
Y component	Free	Rotation Y	0
Z component	0	Rotation Z	0

Table (04)

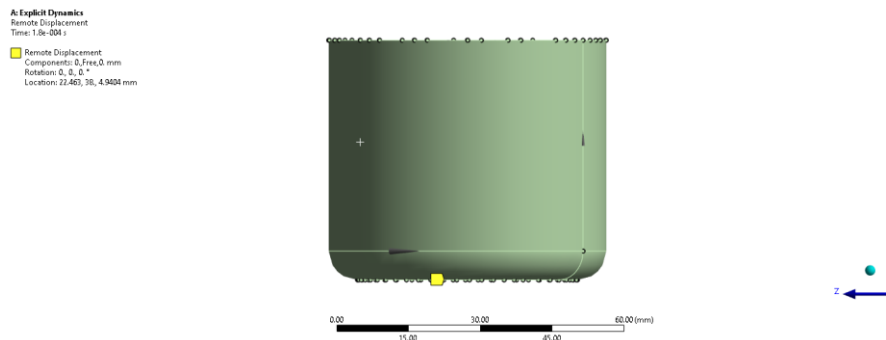


Fig. (6.1.1)

6.2. Die & Blank Holder

A fixed support was set to the die and the blank holder restricting all degrees of freedom. As shown in figure (6.2.1). In real life, the blank holder is not fixed the same way but there's force applied pushing it to the sheet to keep the sheet from wrinkling and control its deformation. The fixed support in the simulation will play the same role as the blank pushes against the blank holder in creating reaction force which keeps the blank in place.

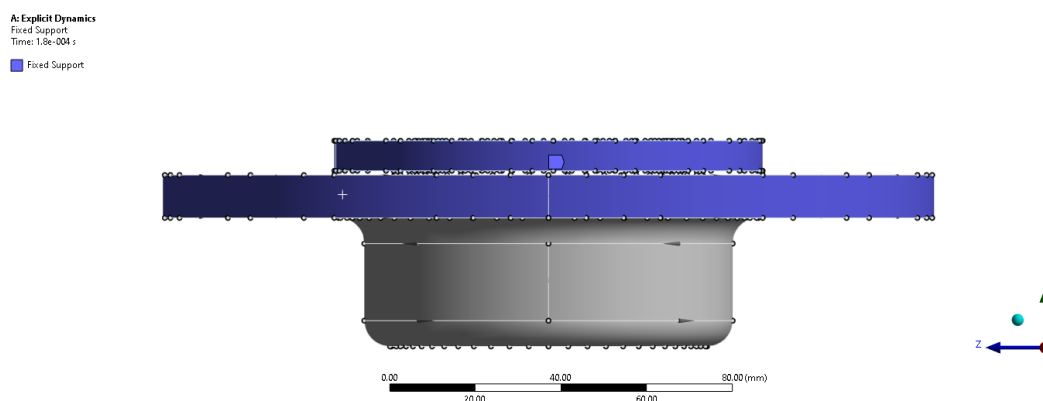


Fig. (6.2.1)

6.3. Blank

Blank was left free between the die and the blank holder without restricting its degrees of freedom. Fixing the blank holder in place is enough to keep it in place, till it takes the shape between the punch and the die.

7. Mesh

7.1. Mesh Convergence

The mesh convergence investigation is important to define how much the results depend on the mesh. The aim is to define a fine mesh to give accurate results but also as rough as possible to save simulation time and money.

Table (05) shows results of the mesh investigation where different meshing values were tried, and the maximum equivalent stress and the simulation time is traced to define the effect of the meshing value. The results showed almost constant stress values between the different mesh settings, but the time significantly increases.

Assuming that the mesh of the die and punch can be set around 3-4 mm, and it doesn't affect the simulation. The mesh dependencies graph with respect to simulation time can be seen in figure (7.1.1), and the mesh dependencies graph with respect to the maximum equivalent stress in figure (7.1.2). The chosen mesh can be the 3rd option as blank will be 2 elements thick for better deformation results while keeping the simulation time acceptable.

Mesh No.	Time [min]	Mesh Size [mm]			Max Stress [MPa]
		Die & Punch	Blank Surface	Blank Thickness	
1	3.9	4	2	2	289.2
2	10.4	4	1	1	288.54
3	14	4	1	0.5	289.82
4	38.5	4	0.5	0.5	289.95
5	65.3	4	1	0.2	289.99

Table (05)

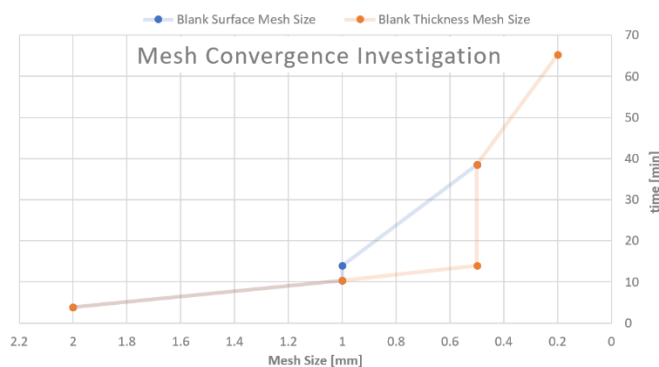


Fig. (3.1.1)

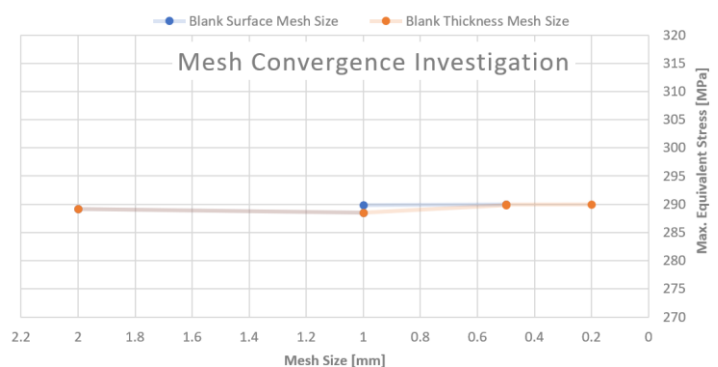


Fig. (3.1.1)

7.2. Final Meshing

The blank mesh is shown in figure (7.2.1). The blank used body meshing of 1 mm and surface meshing of 0.5 mm to create meshing which is 2 elements thick as the blank thickness is 1 mm as shown in figure (7.2.2).

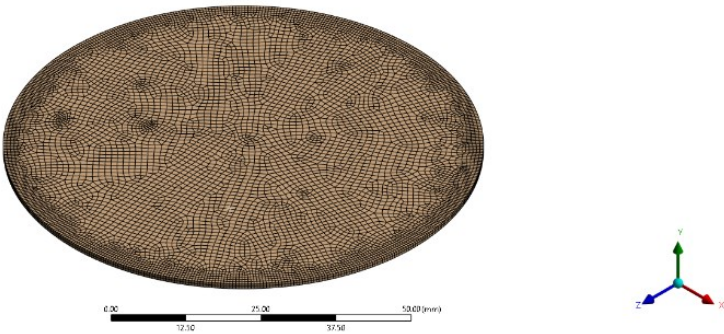


Fig. (7.2.1)

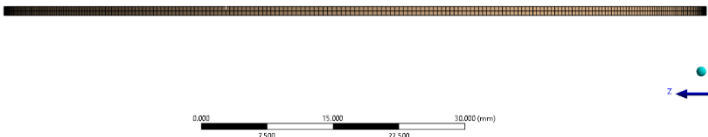


Fig. (7.2.2)

The die, blank holder, and punch used body meshing of 4 mm which is an average medium meshing size for these relatively bigger parts. The mesh is shown in figures (7.2.3) and (7.2.4).

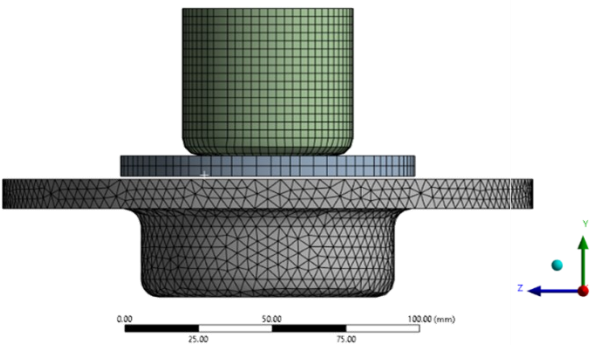


Fig. (7.2.3)

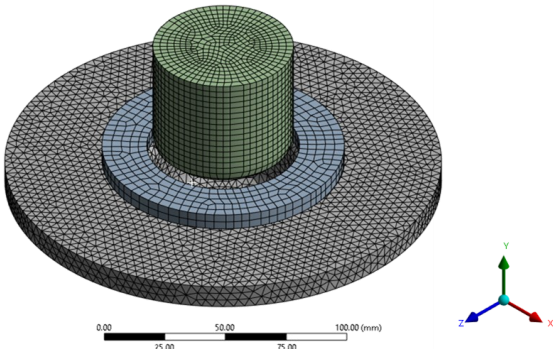


Fig. (7.2.4)

7.3. Mesh Statistics

Table (06) shows mesh statistics values, which includes the total number of elements and total number of nodes. Nodes are the intersection points of the elements where the software calculates the result values of the simulation.

No. of Nodes	46877
No. of Elements	74904

Table (06)

8. Results

8.1. Total Deformation

Fig (8.1.1) shows part of the deep drawing process as the sheet is wrinkling before the punch reshape it back in the correct cylindrical shape.

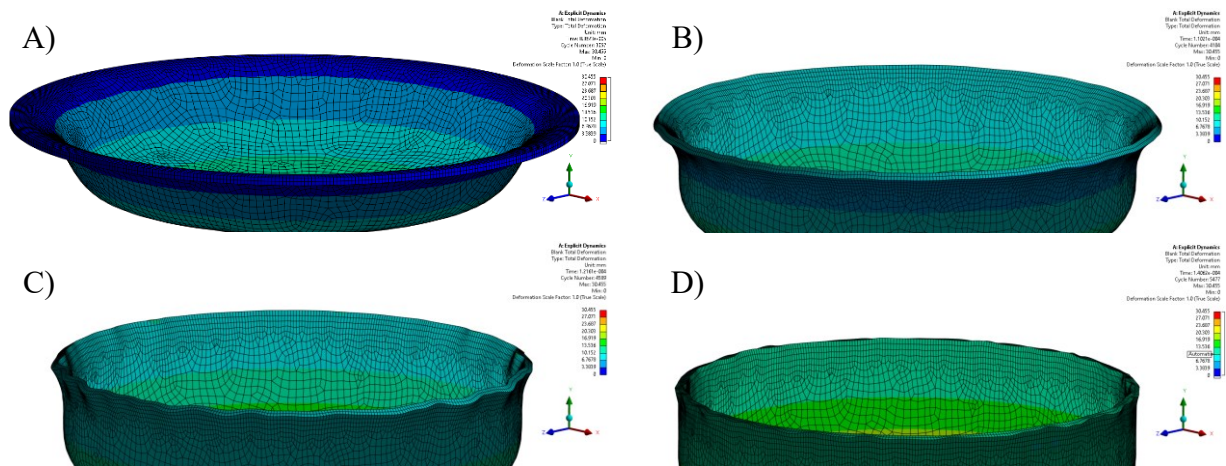


Fig. (8.1.1), A-D

8.2. Equivalent Plastic Strain

Strain can be an indication of the critical failure location. It can also be compared to the material failure strain value to determine if the material has failed. Nonlinear aluminum was used to generate these results. The maximum value of plastic strain is 0.3986 [1] which is less than the failure plastic strain.

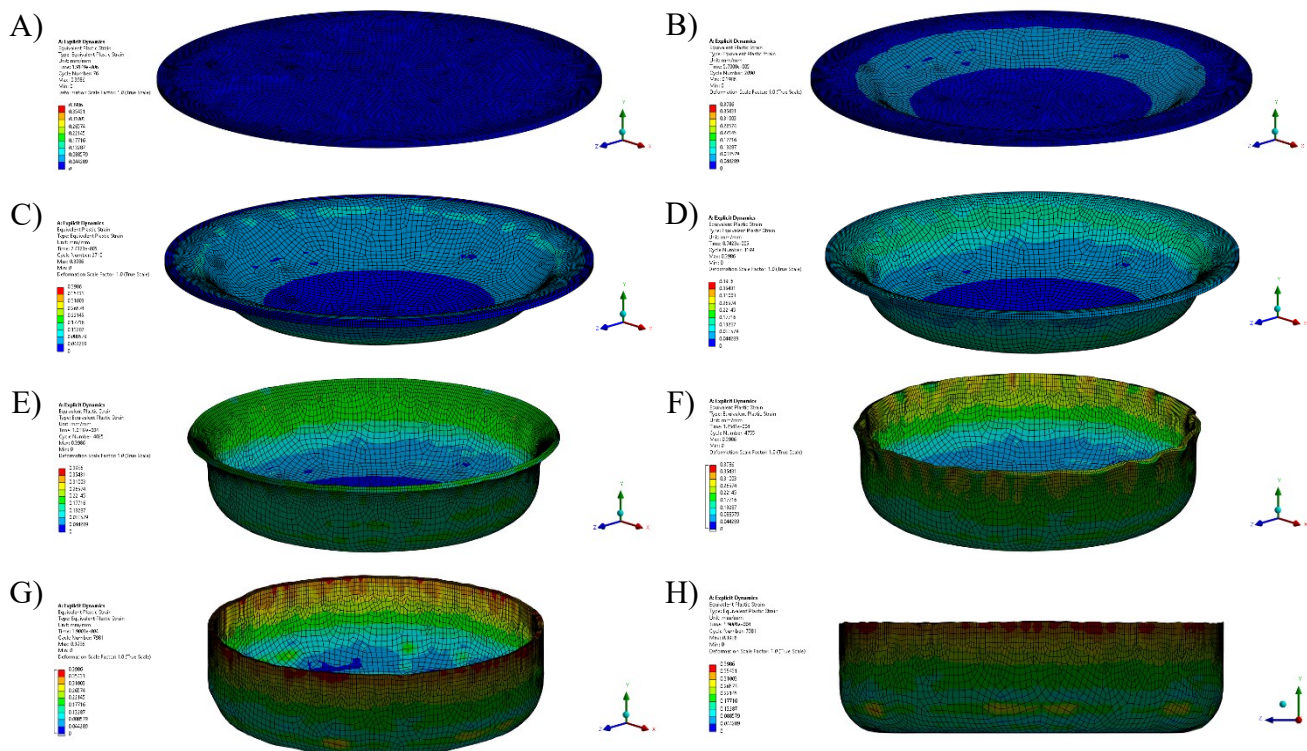


Fig. (8.2.1), A-H

8.3. Equivalent (Von-Mises) Stress

As the stress value varies during the drawing process, Figure (8.3.1) shows the equivalent von-mises stress in different time instants. Maximum stress is 610.66 [MPa]. And the average stress value was around 400 [MPa].

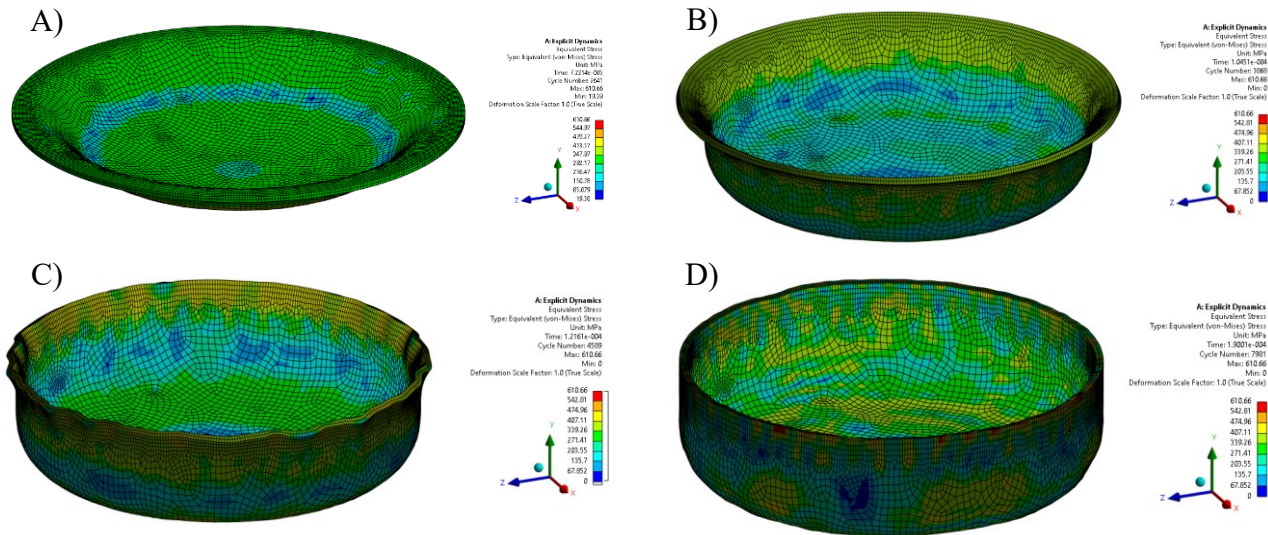


Fig. (8.3.1), A-D

9. Defected Results

9.1. Generating Penetrating the sheet

If the velocity of the punch increases, it can cause cutting the aluminum sheet to be more like the blanking operation instead of the deep drawing operation. Figures (9.1.1) and (9.1.2) show the total deformation of 2 different simulation results. Figure (9.1.3) shows the equivalent plastic strain value during failure which is 1.4923 [1].

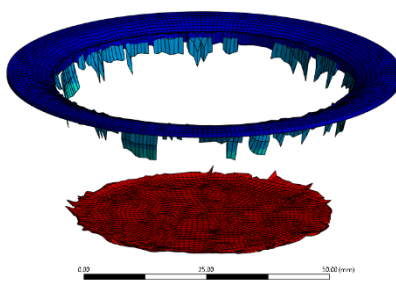


Fig. (9.1.1)

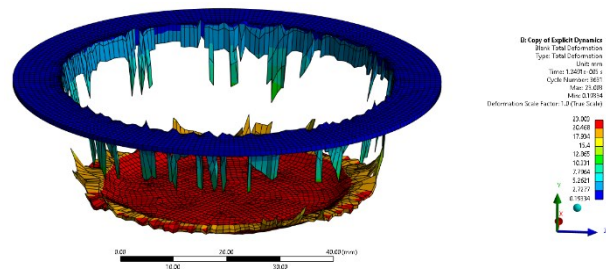


Fig. (9.1.2)

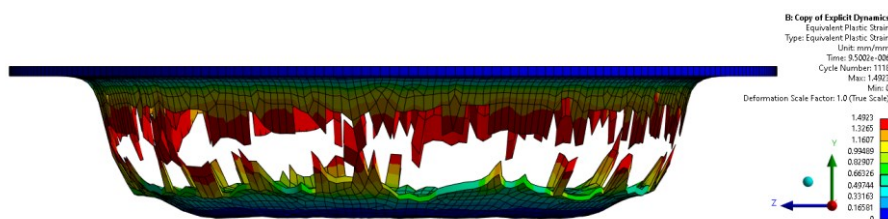


Fig. (9.1.3)

9.2. Tearing

Tearing can also be caused by inappropriate velocity or other parameters resulting in some locations becoming thinner than average or the material locally reaches the plastic strain which results in the failure of the material. Figures (9.2.1) and (9.2.2) show a side view and isometric view of the total deformation of the blank.

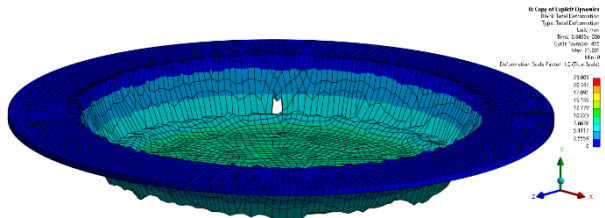


Fig. (9.2.1)

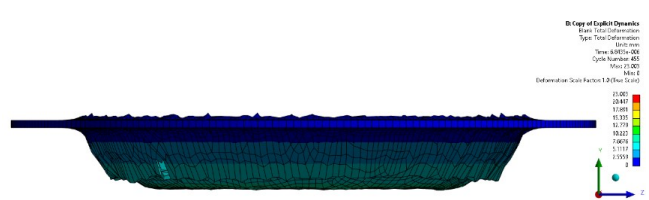


Fig. (9.2.2)

9.3. Earing

The main problem of the earring defect is that the surface needs to be cut of “extra operations needed” to get a regular shape or smooth surface based on the use of the final product. Figure (9.3.1) shows the earring of the cylindrical cup.

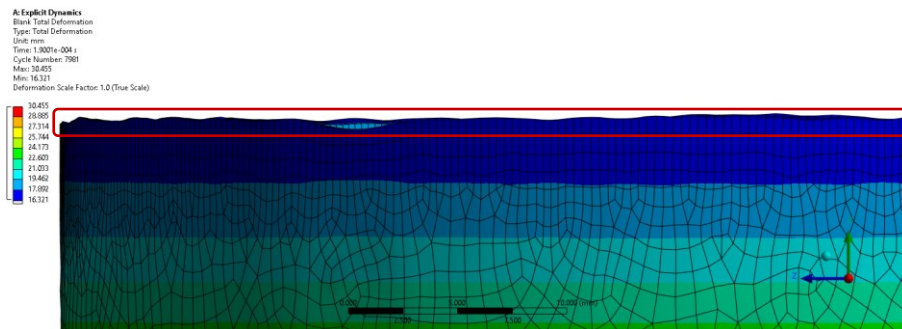


Fig. (9.3.1)

9.4. Wall Wrinkling

The wall wrinkling defect can be caused by geometrical problem of the die as it was generated in this simulation, or any other Error related to it. Figure (9.4.1) shows the total deformation of the blank.

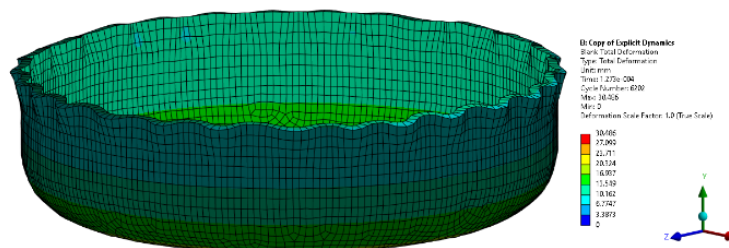


Fig. (9.4.1)